

WEEK 3: SELLER AND INVENTORY MANAGEMENT SYSTEM

Scenario

The **Photo Frame Shopping System** sells photo frames supplied by multiple sellers. The system maintains seller details, tracks the products supplied by each seller, and monitors inventory levels to ensure product availability.

Create Seller and Inventory Tables

Two tables are created for this system:

SELLER – stores seller details such as Seller ID, Seller Name, Email, Phone and Address.

INVENTORY – stores inventory details such as Inventory ID, Product ID, Seller ID, Stock Quantity, Stock Status and Last Updated.

```
SQL> CREATE TABLE SELLER(SELLER_ID VARCHAR(20),SELLER_NAME VARCHAR(10),EMAIL VARCHAR(30),PHONE VARCHAR(20),ADDRESS VARCHAR(50) );  
Table created.
```

```
SQL> CREATE TABLE INVENTORY(INVENTORY_ID VARCHAR(10),PRODUCT_ID VARCHAR(7),SELLER_ID VARCHAR(10),STOCK_QUANTITY NUMBER,STOCK_STATUS VARCHAR(15),LAST_UPDATED  
DATE );  
Table created.
```

Establish Relationships:

Seller and Product :One seller can supply multiple products.One product can be supplied by one or multiple sellers.

Relationship: Seller → Product = One to Many

Product and Inventory: Each product has inventory information. Inventory tracks the available stock.

Relationship: Product → Inventory = One to One

Primary Key:

The following Primary Keys are added:

- **SELLER_ID** – Primary Key of SELLER table.
- **INVENTORY_ID** – Primary Key of INVENTORY table.

A Primary Key uniquely identifies each record in a table.

```
SQL> ALTER TABLE SELLER ADD CONSTRAINT PK_SELLER PRIMARY KEY (SELLER_ID);  
Table altered.  
  
SQL> ALTER TABLE INVENTORY ADD CONSTRAINT PK_INV PRIMARY KEY(INVENTORY_ID);  
Table altered.
```

Foreign Key:

The following Foreign Keys are used in the INVENTORY table:

- **PRODUCT_ID** references PRODUCT(PRODUCT_ID)
- **SELLER_ID** references SELLER(SELLER_ID)

These Foreign Keys establish the relationship between the tables.

```
SQL> ALTER TABLE INVENTORY
2  ADD CONSTRAINT FK_PROD
3  FOREIGN KEY (PRODUCT_ID)
4  REFERENCES PRODUCT(PRODUCT_ID);
```

Table altered.

```
SQL> ALTER TABLE INVENTORY ADD CONSTRAINT FK_SELLER FOREIGN KEY(SELLER_ID) REFERENCES SELLER
(SELLER_ID);
```

Table altered.

Constraints to make unique email ID:

```
SQL> ALTER TABLE SELLER ADD CONSTRAINT UN_MAIL UNIQUE(EMAIL);
```

Table altered.

TO make Email NOT NULL:

```
SQL> ALTER TABLE SELLER MODIFY EMAIL NOT NULL;
```

Table altered.

ADD Constarints to checj the stock Quantity is positive:

```
SQL> ALTER TABLE INVENTORY ADD CONSTRAINT CK_QUAN CHECK(STOCK_QUANTITY>=0);
```

Table altered.

Insert values to seller tables :

```

SQL> INSERT INTO SELLER
  2 VALUES ('S001','VANI ENTERPRISES','VAMI@GMAIL.COM','8765432109','ANNA NAGAR ,CHENNAI');

1 row created.

SQL> INSERT INTO SELLER
  2 VALUES ('S002','SD & CO','SELLERD827@GMAIL.COM','9876543012','62,THIRUMAL NAGER,THANJAVUR');
Enter value for co:
old  2: VALUES ('S002','SD & CO','SELLERD827@GMAIL.COM','9876543012','62,THIRUMAL NAGER,THANJAVUR')
new  2: VALUES ('S002','SD ','SELLERD827@GMAIL.COM','9876543012','62,THIRUMAL NAGER,THANJAVUR')

1 row created.

SQL> INSERT INTO SELLER
  2 VALUES ('S003','A TO Z','ATOZMARKET@GMAIL.COM','7890654312','90,GERODE TOWM,CHENNAI');

1 row created.

SQL>
SQL> INSERT INTO SELLER
  2 VALUES ('S004','DSVV MANUFACTURES','DSVVMANUF142@GAMI.COM','7824003009','23,THILLAI NAGAR,TRICHY');

1 row created.

```

```
SQL> SELECT * FROM SELLER;
```

SELLER_ID	SELLER_NAME	EMAIL
PHONE	ADDRESS	
S001	VANI ENTERPRISES	VAMI@GMAIL.COM
8765432109	ANNA NAGAR ,CHENNAI	
S002	SD	SELLERD827@GMAIL.COM
9876543012	62,THIRUMAL NAGER,THANJAVUR	
S003	A TO Z	ATOZMARKET@GMAIL.COM
7890654312	90,GERODE TOWM,CHENNAI	
S004	DSVV MANUFACTURES	DSVVMANUF142@GAMI.COM
7824003009	23,THILLAI NAGAR,TRICHY	

Insert values to inventory tables:

```

SQL> INSERT INTO INVENTORY
  2 VALUES ('&INVID','&PRODID','&SELL','&STOCK','&STOCK_STATUS',TO_DATE('&LAST_UPDATED','DD-MM-RR'));
Enter value for invid: IVN0001
Enter value for prodid: 101
Enter value for sell: S001
Enter value for stock: 5
Enter value for stock_status: AVAILABLE
Enter value for last_updated: 12-09-26
old  2: VALUES ('&INVID','&PRODID','&SELL','&STOCK','&STOCK_STATUS',TO_DATE('&LAST_UPDATED','DD-MM-RR'))
new  2: VALUES ('IVN0001','101','S001','5','AVAILABLE',TO_DATE('12-09-26','DD-MM-RR'))

1 row created.

```

```

SQL> INSERT INTO INVENTORY
2 VALUES ('&INVID', '&PRODID', '&SELL', '&STOCK', '&STOCK_STATUS', TO_DATE('&LAST_UPDATED', 'DD-MM-RR'));
Enter value for invid: IVN0002
Enter value for prodid: 102
Enter value for sell: S002
Enter value for stock: 5
Enter value for stock_status: AVAILABLE
Enter value for last_updated: 14-SEP-26
old 2: VALUES ('&INVID', '&PRODID', '&SELL', '&STOCK', '&STOCK_STATUS', TO_DATE('&LAST_UPDATED', 'DD-MM-RR'))
new 2: VALUES ('IVN0002', '102', 'S002', '5', 'AVAILABLE', TO_DATE('14-SEP-26', 'DD-MM-RR'))
1 row created.

SQL> INSERT INTO INVENTORY
2 VALUES ('&INVID', '&PRODID', '&SELL', '&STOCK', '&STOCK_STATUS', TO_DATE('&LAST_UPDATED', 'DD-MM-RR'));
Enter value for invid: IVN0003
Enter value for prodid: 103
Enter value for sell: S003
Enter value for stock: 6
Enter value for stock_status: AVAILABLE
Enter value for last_updated: 16-SEP-26
old 2: VALUES ('&INVID', '&PRODID', '&SELL', '&STOCK', '&STOCK_STATUS', TO_DATE('&LAST_UPDATED', 'DD-MM-RR'))
new 2: VALUES ('IVN0003', '103', 'S003', '6', 'AVAILABLE', TO_DATE('16-SEP-26', 'DD-MM-RR'))
1 row created.

SQL> INSERT INTO INVENTORY
2 VALUES ('&INVID', '&PRODID', '&SELL', '&STOCK', '&STOCK_STATUS', TO_DATE('&LAST_UPDATED', 'DD-MM-RR'));
Enter value for invid: IVN0004
Enter value for prodid: 104
Enter value for sell: S004
Enter value for stock: 9
Enter value for stock_status: AVAILABLE
Enter value for last_updated: 10-SEP-26
old 2: VALUES ('&INVID', '&PRODID', '&SELL', '&STOCK', '&STOCK_STATUS', TO_DATE('&LAST_UPDATED', 'DD-MM-RR'))
new 2: VALUES ('IVN0004', '104', 'S004', '9', 'AVAILABLE', TO_DATE('10-SEP-26', 'DD-MM-RR'))
1 row created.

```

```
SQL> SELECT * FROM INVENTORY;
```

INVENTORY_	PRODUCT_ID	SELLER_ID	STOCK_QUANTITY	STOCK_STATUS	LAST_UPDA
IVN0001	101	S001	5	AVAILABLE	12-SEP-26
IVN0002	102	S002	5	AVAILABLE	14-SEP-26
IVN0003	103	S003	6	AVAILABLE	16-SEP-26
IVN0004	104	S004	9	AVAILABLE	10-SEP-26

REPORT 1 : seller -wise product report

This report shows each seller, the number of products supplied by the seller, and the available stock quantity.

```

SQL> SELECT S.SELLER_NAME,
2 COUNT(I.PRODUCT_ID) AS NUMBER_OF_PRODUCTS,
3 SUM(I.STOCK_QUANTITY) AS AVAILABLE_STOCK
4 FROM SELLER S
5 JOIN INVENTORY I
6 ON S.SELLER_ID = I.SELLER_ID
7 WHERE TRIM(I.STOCK_STATUS) = 'AVAILABLE'
8 GROUP BY S.SELLER_NAME;

```

SELLER_NAME	NUMBER_OF_PRODUCTS	AVAILABLE_STOCK
VANI ENTERPRISES	1	5
SD	1	5
DSVV MANUFACTURES	1	9
A TO Z	1	6

REPORT 2: stock Availability report

This report shows each product, its available stock quantity, and its current stock status.

```
SQL> SELECT P.PRODUCT_NAME,
2         I.STOCK_QUANTITY,
3         I.STOCK_STATUS
4 FROM PRODUCT P
5 JOIN INVENTORY I
6 ON P.PRODUCT_ID = I.PRODUCT_ID;
```

PRODUCT_NAME	STOCK_QUANTITY
STOCK_STATUS	
Classic Wooden Frame AVAILABLE	5
Elegant Acrylic Frame AVAILABLE	5
Family Collage Frame AVAILABLE	6

PRODUCT_NAME	STOCK_QUANTITY
STOCK_STATUS	
Decorative Wall Frame AVAILABLE	9

REPORT 3: Inventory analysis

This report analyzes the inventory by finding the total available products, out-of-stock products, highest stocked products, and average inventory quantity.

TOTAL PRODUCTS AVAILABLE:

```
SQL> SELECT COUNT(*) AS TOTAL_AVAILABLE_PRODUCT
2 FROM INVENTORY
3 WHERE STOCK_STATUS = 'AVAILABLE';
```

TOTAL_AVAILABLE_PRODUCT
4

PRODUCTS OUT OF STOCK

```
SQL> SELECT COUNT(*) AS OUTOFSTOCK_PRODUCTS
2 FROM INVENTORY
3 WHERE STOCK_STATUS ='OUT OF STOCK' ;
```

```
OUTOFSTOCK_PRODUCTS
-----
0
```

HIGHTEST STOCKED PRODUCTS

```
SQL> SELECT P.PRODUCT_NAME,I.STOCK_QUANTITY
2 FROM PRODUCT P
3 JOIN INVENTORY I ON P.PRODUCT_ID = I.PRODUCT_ID
4 WHERE I.STOCK_QUANTITY =(SELECT MAX(STOCK_QUANTITY) FROM INVENTORY);
```

```
PRODUCT_NAME                                STOCK_QUANTITY
-----
Decorative Wall Frame                        9
```

AVERAGE INVENTORY QUANTITY

```
SQL> SELECT AVG(STOCK_QUANTITY)AS AVERGE_INVENTORY
2 FROM INVENTORY ;
```

```
AVERGE_INVENTORY
-----
6.25
```